Bon Woong Ku

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SUMMARY

Semiconductor physical design automation expert with **10+ years** delivering **production-grade placement, routing, timing optimization, and multi-die packaging solutions** at advanced nodes (TSMC N3/N2). Currently architecting **interposer routing and heterogeneous integration flows** for Synopsys 3DIC Compiler. Recipient of the **Synopsys TPG Award (2025)** and **IEEE TCAD Donald O. Pederson Best Paper Award (2022)**. Ph.D. in 3D IC physical design with deep expertise in **die-to-die interconnect, power delivery network optimization, and ML-driven design automation**. Driven to build **next-generation AI silicon at scale**.

ACTIVE ROLES

- **ICCAD 2026** Technical Program Committee member
- **DAC 2026** Technical Program Committee member
- **Synopsys 3DIC Compiler** — Lead R&D engineer for advanced multi-die routing and chiplet integration

CORE EXPERTISE

Multi-Die & Advanced Packaging

Synopsys 3DIC Compiler (2024–present) · Synopsys StarRC (2016) · IMEC (2015–2016) · Georgia Tech Ph.D. (2014–2019)

- Architecting **interposer routing** and **feasibility-aware chiplet integration**, shipped to 10+ customers with 10× exploration speedup at 3DIC Compiler
- Developed **tier-partitioning algorithms** and full-chip physical design flows for gate-level **monolithic 3D ICs** at IMEC
- Built **extraction–PDN co-optimization** flows for 3D integration at StarRC
- Designed **heterogeneous 3D AI accelerator architectures** integrating logic and emerging memory technologies at DARPA CHIPS project
- Created scalable **3D IC methodologies** validated on 10+ designs; developed experimental **PDKs at 10nm/7nm** for PPA analysis at Georgia Tech

Physical Design & Signoff at Advanced Nodes

Synopsys PrimeTime ECO / PrimeECO / PrimeClosure (2019–2024)

- Shipped **timing-aware ECO legalization** with **timer–router–legalizer co-optimization**, improving QoR by 20% across 10+ customer signoff flows, **certified for TSMC N3/N2**
- Led signoff timer integration for Tweaker ECO merge, ensuring first-silicon-success at the most advanced process nodes

AI-Driven Design Automation

Intel Labs (2019) · Georgia Tech Ph.D. (2014–2019)

- Automated standard cell routing at advanced Intel nodes, replacing manual heuristics with **reinforcement learning** — demonstrating ML-guided physical design for high-performance compute silicon
- Developed **ML-based wire parasitic prediction** for monolithic 3D ICs, achieving 98%+ RC accuracy and up to 16× timing improvement in full-chip optimization

Skills

- **C/C++:** production EDA engines — placement, routing, legalization, parasitic extraction, signoff optimization
- **Tcl/Shell:** physical design methodology integration, customer-facing scripting
- **Rust:** systems programming for next-generation EDA tooling
- **Python / PyTorch:** ML pipelines, design automation for physical design

ACHIEVEMENTS

Honors

- **Technology & Product Development Group Award** — Shankar Krishnamoorthy (CPDO, Synopsys, 2025)
- **Donald O. Pederson Best Paper Award** — IEEE TCAD (2022)
- **Team Award** — Jacob Avidan (SVP, Synopsys, 2022)
- **Contribution Aware, Customer Success Award** — Jacob Avidan (SVP, Synopsys, 2021)
- **Signoff Platform & Innovation Recognition Award** — Jacob Avidan (SVP, Synopsys, 2020)
- **Best Paper Award Nomination** — ACM ISPD (2018)
- **Presidential Science Scholarship** — Korea Student Aid Foundation (2008–2014)

Patent

- **B. W. Ku**, N. Oh, C. Moon. “Timing-aware Surgical Optimization for Engineering Change Order in Chip Design.” U.S. Patent No. 12,175,181, granted Dec. 2024.

Selected Publications

26 total publications · 6 papers at DAC/ICCAD · full list upon request

- **B. W. Ku**, K. Chang, S. K. Lim, “Compact-2D: A Physical Design Methodology to Build Two-Tier Gate-level 3D ICs,” **IEEE TCAD**, 2020 — *Donald O. Pederson Best Paper Award*
- A. Mallik, A. Vandooren, L. Witters, A. Walke, J. Franco, Y. Sherazi, P. Weckx, D. Yakimets, M. Bardon, B. Parvais, P. Debacker, **B. W. Ku**, S. K. Lim, A. Mocuta, D. Mocuta, J. Ryckaert, N. Collaert, and P. Raghavan, “The Impact of Sequential-3D Integration on Semiconductor Scaling Roadmap,” **IEDM**, 2017

EDUCATION

Georgia Institute of Technology — Ph.D. & M.S., Electrical and Computer Engineering, 2019
Thesis: Physical Design Solutions for 3D ICs and their Neuromorphic Applications

Seoul National University — B.S., Electrical and Computer Engineering, 2014